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ENGR 095

September 9 2026

Reflecting on the Bhopal Disaster

The Bhopal disaster occurred at a Union Carbide plant making pesticides in 1984. Poisonous gas was released into a highly populated area in India, resulting in tragic loss of life. The plant, which opened in 1978, produced the chemical MIC in its liquid form as a precursor to other chemicals. Multiple earlier incidents foreshadowed the tragedy to come. In 1981, a maintenance worker was accidentally exposed to MIC gas and did not survive. An employee of the chemical plant raised concerns about safety and lack of procedures for evacuations and knowledge of danger, but was ignored by managers. The actual leak occurred when a worker conducted maintenance improperly, allowing water to seep into liquid MIC tanks and begin reacting to create a gas. The plant had multiple safety systems to prevent a leak, but none were activated, possibly due to budget cuts. Toxic gas eventually escaped the plant and blew into the unprepared city of Bhopal. The chemical plant did not notify local authorities that anything had gone wrong, and had made no effort to create evacuation procedures for the event of a leak. This left civilians with no information, and many died trying to run from the gas or shelter in their homes. Local hospitals were overrun and had no information about how to treat the symptoms of exposure. The exact death toll remains unknown, but some estimates put it at over ten thousand people. In the aftermath of the disaster, Union Carbide made efforts to downplay its role and avoid responsibility.

This disaster forces us to confront some difficult questions about the nature of

technological progress and its interaction with developing nations. The fact that this disaster happened in India is of vital importance to understanding it. We must ask ourselves: why do we seek to outsource these types of factories overseas? Are we bringing jobs and prosperity to people in desperate need, or are we placing them at an undue risk and exploiting lower wages and laxer regulations? Beyond those concerns, who is at fault for what happened? Was it the maintenance worker who followed the procedure incorrectly? Was it Union Carbide, who allowed unsafe conditions to persist? Was it the operators, who decided to disable the safety systems?

The truth is, none of these questions have easy answers. When it comes to placing blame, it is perhaps the case that all of those listed share some fault, but it is undeniable that the way the government and the company designed the system around the plant enabled the disaster. This tragedy did not occur solely because of one mistake. It occurred because all it took to kill tens of thousands was one small mistake. That is a systemic failure. It required a series of decisions by many people. Most surely had no intention of causing a disaster, but their own individual needs and circumstances took priority over the people they had a duty to protect. The conditions of the plant's location in India are likewise complex. It is true that jobs at the plant provided upward mobility for many families in Bhopal. It fostered a sense of community and pride. But it is also true that Union Carbide benefitted from the lower labor costs of manufacturing in India, and were perhaps more neglectful. At a similar plant in West Virginia, the company maintained stricter oversight and more sophisticated monitoring systems.

Having now reflected on multiple engineering disasters, I observe that most share two common root causes: greed and hubris. The greed of the Union Carbide company led them to build a plant in a populated area overseas, and cut costs on safety measures. The hubris of the

operators and engineers working at the plant led them to assume that nothing would go wrong. They were too proud to create evacuation procedures or treatment plans, as they assumed they would never be necessary. As engineers, it is our responsibility to consider both all of the ways a system can fail, and the impacts of that failure on the surrounding world. We must not assume ourselves infallible, but create checks and systems that mitigate the impact of one small mistake. A maintenance worker doing their job improperly should never be able to result in a disaster of this magnitude. We also have a responsibility to society. If we feel that we are creating a system that is unsafe, due to time or cost pressures, it is our duty to refuse to create that system, and warn the public of its existence.